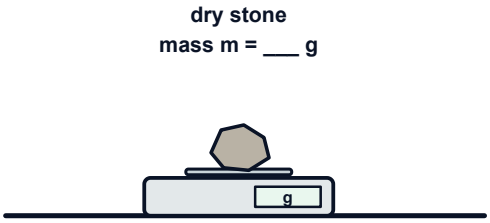
DO NOW · REQUIRED PRACTICAL

Density of an irregular solid

Particle model of matter
Paper 1 · Combined and Separate
6 steps

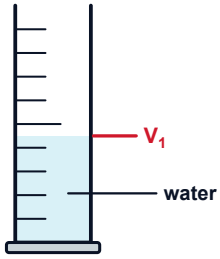
A student measures the density of a small stone.

1 Measure the mass of the dry stone with a balance.



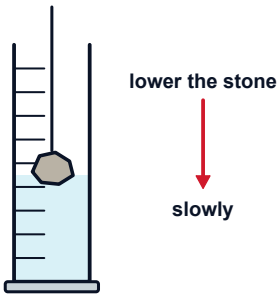
measure the mass

2 Part-fill a measuring cylinder with water and record the volume.

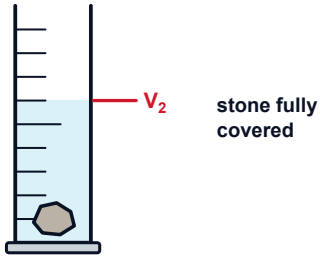


part-fill with water and record the volume V_1

3 Lower the stone into the water until it is fully covered.



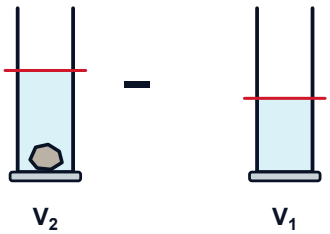
4 Record the new volume shown on the measuring cylinder.



record the new volume V_2

5 Subtract the two volumes to find the volume of the stone.

volume of stone = $V_2 - V_1$



6 Calculate the density: mass ÷ volume.

$$\rho = m \div V$$

density (g/cm³) = mass (g) ÷ volume (cm³)

m = mass of the stone, V = $V_2 - V_1$

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